Skills Summary

Dividing Complex Numbers with Conjugates

Use this reference while completing your worksheet or when studying.

A quotient is not finished until the denominator is real. The conjugate of $a+bi$ is $a-bi$: same real part, opposite imaginary part. Multiplying top and bottom by it never changes the value, only the form.

1. Why the conjugate clears a denominator

Rule: $(a+bi)(a-bi) = a^2 - b^2i^2 = a^2 + b^2$, a real number. The outer and inner products cancel, and the $-b^2i^2$ becomes $+b^2$. So the conjugate is the one multiplier guaranteed to make a denominator real.

Example: Multiply $6+5i$ by its conjugate.
Step 1. The conjugate flips the sign of the imaginary part only, so it is $6-5i$; the point is to watch the i terms disappear.
Step 2. $(6+5i)(6-5i) = 36 - 30i + 30i - 25i^2 = 36 + 25$.
61

Try it: Multiply $3+7i$ by its conjugate.

check: **58**

2. Dividing by a pure imaginary number

Rule: the conjugate of bi is $-bi$, and $(bi)(-bi) = b^2$. Useful shortcut: $\dfrac{1}{i} = -i$, so dividing by i is multiplying by $-i$. Write the answer as $a+bi$, real part first.

Example: Simplify $\dfrac{10+15i}{5i}$.
Step 1. The denominator has no real part, so its conjugate is just $-5i$ — one multiplication clears it.
Step 2. $\dfrac{(10+15i)(-5i)}{(5i)(-5i)} = \dfrac{-50i+75}{25} = \dfrac{75-50i}{25}$.
$\boldsymbol{3-2i}$

Try it: Simplify $\dfrac{8-2i}{2i}$.

check: $\boldsymbol{-1-4i}$

3. Dividing by a full complex denominator

Rule: multiply top and bottom by the conjugate of the denominator, then divide *both* parts of the new numerator by the real denominator: $\dfrac{z}{a+bi} = \dfrac{z(a-bi)}{a^2+b^2}$. Dividing real by real and imaginary by imaginary is not a legal move.

Example: Simplify $\dfrac{11+3i}{3+2i}$.
Step 1. The denominator is a full binomial, so the conjugate $3-2i$ is what turns the bottom into the real number $9+4$.
Step 2. $\dfrac{(11+3i)(3-2i)}{13} = \dfrac{33-22i+9i+6}{13} = \dfrac{39-13i}{13}$.
$\boldsymbol{3-i}$

Try it: Simplify $\dfrac{5+5i}{1+3i}$.

check: $\boldsymbol{2-i}$

4. Circuits: voltage, current and impedance

Model: in an AC circuit $V = I \cdot Z$, where voltage V is in volts, current I in amps and impedance Z in ohms — all three complex. So $Z = V/I$ and $I = V/Z$: every missing quantity is one division by the quantity you already know.

Example: $V = 1+7i$ volts across an impedance of $1+2i$ ohms. Find the current.
Step 1. Current is wanted and V and Z are known, so rearrange $V = IZ$ to $I = V/Z$ — a division by the impedance.
Step 2. $\dfrac{(1+7i)(1-2i)}{1+4} = \dfrac{1-2i+7i+14}{5} = \dfrac{15+5i}{5}$.
$\boldsymbol{3+i}$ amps

Try it: $V = 4+7i$ volts, $Z = 2+i$ ohms. Find the current in amps.

check: $\boldsymbol{3+2i}$